



FIRST SEMESTER 2026-2027
COURSE HANDOUT

Date: 01.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : MEL G 628
Course Title : Digital VLSI Design
Instructor-in- Charge : Dr. Nitin Chaturvedi
Lab Instructors : Mr. Shubham
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1. Course Description:

We will study the fundamental structures of VLSI Systems at the lowest levels of system abstraction, namely those associated with the direct application of VLSI devices to particular problems of interest. The broad topics of coverage includes Principles of operations of CMOS transistors used as a digital switch and various other topologies used in Digital VLSI Design, Logic implementation strategies and performance characterization of the VLSI circuits, Low Power Design, different clocking strategies, symbolic layout systems, CMOS subsystems design and Memory modules.

2. Scope and Objective:

The objective of this course is to understand the need and importance of digital VLSI design in CMOS integrated circuits. To understand different design principles used in the design of Digital VLSI Circuits & Systems using CMOS with emphasis on low power and high performance.

3. Text Book:

a).(T1) Essentials of VLSI Circuits and Systems, Kamran Eshraghian, Douglas A. Pucknell and Sholeh Eshraghian, PHI, 2005/2009.

b).(T2) CMOS Digital Integrated Circuit, Analysis and Design, Sung-Mo Kang and Yusuf Leblebici, Publisher: McGraw- Hill Companies, Inc.2003

4. Reference Books:

a).(R1) Digital Integrated Circuits, A Design Perspective, Jan M Rabaey, Anantha Chandrakasan, Borivoje Nikolic, 2nd edition, Prentice Hall, 2005.



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5. Course Plan :

Module	Lect Session	Topics to be covered	Reference to Text	Learning Outcome
1	1	Introduction to VLSI Design	Chapter-1 (T2)	Learn: VLSI Design Methodologies
1	2	Introduction to MOS Physics	2.2(T1), 3.1, 3.2, 3.3(T2)	Learn: Basic Electrical Properties of MOS Device
1	3	Device Characteristics	2.1, 2.3(T1);3.4(T2)	Knowledge of : Drain-to-Source Current versus Voltage relationships
2	4-6	Static NMOS/CMOS inverter: voltage-transfer characteristics	2.4; 2.6; 2.10 (T1), Chapter 4 (T2), Chapter 5 (R1)	Learn: Voltage Transfer Characteristics of CMOS Inverters; Performance measure of CMOS Inverters
2	7-10	MOS Capacitances, Dynamic Behavior of CMOS Inverters	4.6;4.7(T1) Chapter 6 (T2)	Learn: Capacitances associated with CMOS devices and switching characteristics of CMOS inverters
3	11-13	CMOS Process Technology, Layout and Design Rules	Chapter 3 (T1)	Learn: CMOS Fabrication steps, Stick Diagrams and Layout Design rules
3	14	Scaling of MOS Devices	Chapter 5 (T1)	Learn: Constant Voltage Scaling; Full Scaling; Short channel and narrow channel effects
3	15-20	Combinational CMOS logic circuit	6.2-6.4,(T1) Chapter 7 (T2);	Learn: Basic building blocks for combinational logic design, optimisation for speed, method of logical effort, Euler method, Ratioed logic, pseudo-NMOS logic, Pass transistor logic, DCVSL
3	21	InterConnect Parasitics	4.2-4.10(T1) 6.5-6.7 (T2)	Learn: Understanding Interconnect, Capacitances, Resistance, Delay models
4	22-24	CMOS Sequential logic Design	6.5-6.6(T1) Chapter 7(R1)	Learn: Design of Flip-flops, Latches & registers
4	25-28	Dynamic & Domino Logic Design	6.6 (T1) Chapter 7(R1) Chapter 9 (T2)	Learn: Dynamic CMOS Circuit techniques, dynamic latches & registers; NORA logic
4	29-30	Clocking Strategies	6.6 (T1) Chapter 7(R1) Chapter 9 (T2)	Learn: Synchronous design and timing metric
5	31-34	Aspects of Memory design	Chapter 9 (T1); Chapter 10(T2)	Learn: Design of SRAM, DRAM, decoders, sense amplifiers
5	35-36	Adder architectures	8.4-8.5(T1); Chapter 11(R1)	Learn: Tx level implementation of Adders, Multipliers & Shifter
5	37-40	Introduction to Finfet based circuit design	Study Material	Learn: Finfet circuit design



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6. Lab Plan:

SL. No	Topics	No of Cycles (2hr/cycle)
1	Circuit Design & Simulation (spice code)	2
2	Schematic Entry (Simulation)	2
3	Circuit Extraction	1
4	Layout Design	2
5	DRC, LVS, Parasitic extraction	2
6	Post Layout Simulation	1

7. Evaluation Scheme :

SL. No	Component & Nature	Weightage	Duration	Date	Nature
1.	Labs Assignments	20 (20%)	Continuous	Weekly	OB
2.	Mid-Semester Exam	25 (25%)	90 min	7/10/2026 FN2	CB
3.	Design Projects+Seminar (Research Paper/Report Writing)	20 (20%)	15-20 days	10/11/2026	OB
4.	Comprehensive Exam	35 (35%)	3 hrs	08/12/2026 AN	CB

8. Chamber Consultation Hour: Thursday: 6pm to 7pm.

9. Make-up Policy: No make up for Lab Assignments & Project.

10. Course Notice: Will be displayed on Nalanda Notice board/Direct Email.

Note: A student who scores less than 20% marks will be awarded NC.

Instructor-in-Charge
MEL G 628